

Listening in the Dark: The sensory ecology of Neotropical electric fishes

Mon, November 9th
12:00 – 1:30 PM
RB 130

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Neotropical electric fishes use self-generated electric fields to sense their surroundings and communicate in darkness. Their signals provide a window into how sensory systems shape ecological interactions and evolutionary diversification. This talk will introduce the diversity and natural history of these fishes, then present two studies combining field observations with experimental and comparative approaches. First, recordings from Amazonian streams reveal how electric eels and their knifefish prey suppress their signals during hunting and predator avoidance, balancing the benefits of electrolocation against the costs of revealing their presence. Second, a continental study of hypopomid knifefishes shows that coexisting species collectively have more distinct electric signals than expected by chance. The talk will conclude by considering what these patterns suggest about species recognition, signal evolution, and the assembly of diverse tropical fish communities.